INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

Infosys SE Complete Preparation — Divisibility Rules & Remainder Theorems

Section: prepinsta infosys SE | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

Infosys SE Complete Preparation — Divisibility Rules & Remainder Theorems

Source Portal: PrepInsta Infosys Preparation Suite (`https://prepinsta.com/infosys/aptitude/divisibility/`)
Target Role: Infosys Systems Engineer (SE) | **Category:** 01_Mathematical_Ability_Technical_Aptitude
Total Verified Questions Extracted: 10
Quizzes Covered: quiz-1, quiz-2, quiz-3, quiz-4

📋 Section 1: Rules to Keep in Mind

These rules and fundamental principles govern all Infosys questions on this topic:

- Divisibility by 2: Last digit is even; by 4: Last 2 digits divisible by 4; by 8: Last 3 digits divisible by 8.
- Divisibility by 3: Sum of digits divisible by 3; by 9: Sum of digits divisible by 9.
- Divisibility by 5: Last digit 0 or 5; by 10: Last digit 0.
- Divisibility by 11: Difference between sum of odd-positioned digits and even-positioned digits is 0 or multiple of 11.
- Divisibility by composite numbers (e.g. 72): Number must be divisible by co-prime factors 8 and 9.

⚡ Section 2: How to Solve Quickly

Proven time-saving strategies recommended by PrepInsta for Infosys 2026 examination:

1. **Option Elimination via Unit Digits & Multiples:** Before doing full calculations, inspect the options to see if last digits or divisibility by 3, 7, 9, 11 can eliminate 2 to 3 wrong choices.
2. **Approximation & Magnitude Checks:** When exact arithmetic is tedious, round numbers to neat multiples (e.g., 99 ≈ 100) to find the nearest valid option.
3. **Reverse Option Substitution:** In complex algebraic equations, plug option values directly into the question stem to test validity.

🚀 Section 3: Tricks & Shortcuts

Key mathematical formulas, mental-math shortcuts, and speed axioms:

- Wilson's & Fermat's Little Theorem: a^(p-1) mod p = 1 where p is prime and gcd(a, p) = 1.

- Unit digit of large powers cyclicity: 2, 3, 7, 8 have cyclicity 4; 4, 9 have cyclicity 2; 0, 1, 5, 6 have cyclicity 1.

📚 Section 4: Practice Questions, Options & Step-by-Step Solutions

Contains **10** verbatim questions with options, verified answer keys, and complete derivations:

Question 1 (PrepInsta quiz-1 - Q1)

Question Stem:
Which digit should come in the place of * and # if the number 62684*# is divisible by both 8 and 5 ?

Options:
- **(A)** `4 and 0`
- **(B)** `0 and 5`
- **(C)** `5 and 0`
- **(D)** `2 and 0`

Correct Answer: (A) `4 and 0`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Since the given number is divible by 5,So 0 or 5 must come in place of * .But a number ending with 5 is never divisible by 8.Therefore 0 will be in units place


The number formed by the last 4 digit is 4 *0which becomes divisible by 8, if * is replaced by 4.
```

Question 2 (PrepInsta quiz-1 - Q2)

Question Stem:
Find the total number of prime factors in the expression 4 11 ×7 5 ×11 2

Options:
- **(A)** `18`
- **(B)** `31`
- **(C)** `29`
- **(D)** `20`

Correct Answer: (C) `29`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
((2)2)11 × 7 5  × 112=222 × 75× 112


Total number of prime factors =22+5+2=29
```

Question 3 (PrepInsta quiz-1 - Q3)

Question Stem:
If an integer n is divisible by 3, 5 and 12, what is the next larger integer divisible by all these numbers?

Options:
- **(A)** `n + 3`
- **(B)** `n + 5`
- **(C)** `n + 12`
- **(D)** `n + 60`
- **(E)** `n + 15`

Correct Answer: (D) `n + 60`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
n is divisible by 3, 5 and 12 it must a multiple of the lcm of 3, 5 and 12 which is 60.


n = 60 k


n + 60 is also divisible by 60 since


n + 60 = 60 k + 60 = 60(k + 1)


Video solution:
```

Question 4 (PrepInsta quiz-1 - Q4)

Question Stem:
When the integer n is divided by 8, the remainder is 3. What is the remainder if 6n is divided by 8?

Options:
- **(A)** `1`
- **(B)** `2`
- **(C)** `3`
- **(D)** `4`

Correct Answer: (B) `2`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
When n is divided by 8, the remainder is 3 may be written as

n = 8 k + 3

multiply all terms by 6

6 n = 6(8 k + 3) = 8(6k) + 18

Write 18 as 16 + 2 since 16 = 8 * 2.

= 8(6k) + 16 + 2

Factor 8 out.

= 8(6k + 2) + 2

The above indicates that if 6n is divided by 8, the remainder is 2. The answer is B.
```

Question 5 (PrepInsta quiz-1 - Q5)

Question Stem:
If n is an integer, when (2n + 2) 2 is divided by 4 the remainder is

Options:
- **(A)** `Zero`
- **(B)** `2`
- **(C)** `3`
- **(D)** `4`

Correct Answer: (A) `Zero`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
We first expand (2n + 2)2

(2n + 2)2 = 4n 2 + 8 n + 4

Factor 4 out.

= 4(n 2 + 2n + 1)

(2n + 2)2 is divisible by 4 and the remainder is equal to 0. The answer is A.
```

Question 6 (PrepInsta quiz-1 - Q6)

Question Stem:
If a positive integer n is divided by 5, the remainder is 3. Which of the numbers below yields a remainder of 0 when it is divided by 5?

Options:
- **(A)** `n + 3`
- **(B)** `n + 2`
- **(C)** `n - 1`
- **(D)** `n - 2`

Correct Answer: (B) `n + 2`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
divided by 5 yields a remainder equal to 3 is written as follows

n = 5 k + 3 , where k is an integer.

add 2 to both sides of the above equation to obtain

n + 2 = 5 k + 5 = 5(k + 1)

The above suggests that n + 2 divided by 5 yields a remainder equal to zero.

Video solution:
```

Question 7 (PrepInsta quiz-1 - Q7)

Question Stem:
The integers 34041 and 32506 when divided by a three-digit integer n leave the same remainder. What can be

the value of N?

Options:
- **(A)** `289`
- **(B)** `367`
- **(C)** `453`
- **(D)** `307`

Correct Answer: (D) `307`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
The difference of the numbers = 34041 – 32506 = 1535


Factors of 1535:-


1, 5, 307, 1535


Here, 307 is a three digit factor of 1535. So N = 307
```

Question 8 (PrepInsta quiz-1 - Q8)

Question Stem:
How many even integers n, where $100 \leq n \leq 200$, are divisible neither by seven nor by nine?...

Options:
- **(A)** `40`
- **(B)** `37`
- **(C)** `39`
- **(D)** `38`

Correct Answer: (C) `39`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Between 100 and 200 both included there are 51 even nos. There are 7 even nos which are divisible by 7 and 6 nos which are divisible by 9 and 1 no divisible by both. hence in total 51 – (7+6)-1 = 39...
```

Question 9 (PrepInsta quiz-1 - Q9)

Question Stem:
Let N = 1421 * 1423 * 1425. What is the remainder when N is divided by 12?

Options:
- **(A)** `4`
- **(B)** `9`

- **(C)** `3`
- **(D)** `6`

Correct Answer: (C) `3`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
The numbers 1421, 1423 and 1425 when divided by 12 give remainder 5, 7 and 9 respectively. 5*7*9
mod 12 = 11 * 9 mod 12 = 99 mod 12 = 3
```

Question 10 (PrepInsta quiz-1 - Q10)

Question Stem:

The rightmost non-zero digit of the number 30 2720 is...

Options:

- **(A)** `1`
- **(B)** `3`
- **(C)** `7`
- **(D)** `9`

Correct Answer: (A) `1`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Rightmost non-zero digit of 302720 is same as rightmost non-zero digit of 3272. 272 is of the form
...


All 34k  end in 1.


Right most non-zero digit is 1....
```
